UNIVERSITY OF WOLLONGONG AUSTRALIA

Tutorial 03

Transfer functions, signal paths and model meaning

Preparation: Lecture 03: zero-state transfer functions, block-diagram algebra and signal definitions; use Lecture 02A for inversions.

Submission credit: 1% of the course grade for submitted work; correctness is not marked.

Show essential balances, substitutions and units.

Question 1 Quick checks on transfer functions and poles

Show a brief reason or calculation for each answer.

a) ODE to transfer function. With zero initial conditions, find X(s)/F(s):

$$2\ddot{x}(t) + 6\dot{x}(t) + 4x(t) = f(t)$$

b) Poles and finite zeros. Identify their number and location for:

$$G(s) = \frac{2(s + 3)}{(s + 1)(s + 4)}$$

c) Stability. Which systems are BIBO stable? Give a reason for each:

$$G_1(s) = \frac{1}{s + 2}, \qquad G_2(s) = \frac{1}{s - 2}, \qquad G_3(s) = \frac{1}{s^2 + 4}$$

d) Negative feedback. With G(s) below and H(s) = 1, find Y(s)/R(s) and its pole:

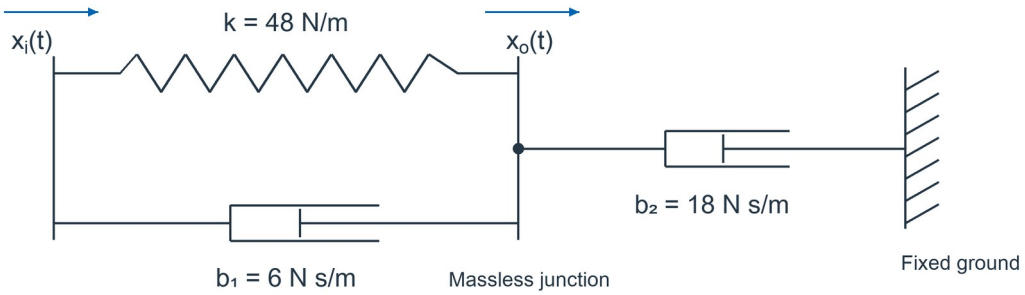
$$G(s) = \frac{2}{s + 1}$$

e) System and input poles. Is the origin a system pole? Find the step final value:

$$G(s) = \frac{1}{s + 2}, \qquad R(s) = \frac{1}{s}, \qquad Y(s) = \frac{1}{s(s + 2)}$$

Question 2 A massless viscoelastic mounting network

An imposed input displacement xi(t) is connected to an output junction xo(t) by a spring k = 48 N/m and damper b_1 = 6 N s/m in parallel. A second damper b_2 = 18 N s/m connects the output junction to ground. The junction is idealised as massless. Use rightward displacement as positive.



a) Balance forces at the junction and derive the zero-state displacement transfer function Xo/Xi.

b) Identify the pole, zero, DC gain and high-frequency gain.

Question 3 Block diagrams and disturbance transfer function

A normalized process is defined by $E=R-Y-0.5V$, $V=G_1E$, and $Y=G_2(V+D)$. The internal signal V is taken before the disturbance junction. D is added at the input to G_2. All initial states are zero.

$$G_1(s)=\frac{6}{s+3},\ G_2(s)=\frac{2}{s+5}$$

a) Draw the complete block diagram from these signal equations. Label R, E, V, D and Y; show both negative feedback paths and the positive disturbance input.

b) Set R = 0. Simplify the inner feedback loop, then draw the reduced block diagram from D to Y and find Y(s)/D(s). Keep both feedback paths connected. Leave the denominator unexpanded.
MATLAB verification. After simplifying by hand, use tf and feedback to check the inner loop and D-to-Y transfer function. Report numerator and denominator coefficients. Control System Toolbox is required.

Question 4 Free motion is not a transfer function

A new plant satisfies $2x'' + 7x' + 3x = f(t)$, with $x(0) = 0.100$ m and $x'(0) = -0.200$ m/s.

a) Find displacement and velocity transfer functions under zero initial conditions.

b) Write the full $X(s)$ including initial terms. Set $f = 0$, invert the free response and check its initial values.

c) Explain why dividing the total free-response transform by F is not a plant transfer function.

Method reference

Method source: Ogata, System Dynamics, 4th ed., Problem A-4-2, pp. 136–137 (massless spring–damper network); Modern Control Engineering, 5th ed., Chapter 2 (block diagrams). The network output and nested-loop task are independently adapted. Page numbers refer to printed pages.

Prof. Zengxi (Stephen) Pan | University of Wollongong | TYUT 2026